Failure Analysis Report

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Inspection Details

Equipment	EQ-INSP-01	Lot	LOT-2024-COLAB
Wafer	W05	Modality	sem

Severity

CRITICAL | Yield impact: 1.0% | Justification: The crack is small in size and does not appear to be directly affecting any critical components or pathways within the chip. It may have minimal effect on performance if it remains stable over time. However, depending on its location and potential for growth, further investigation may

Defect Description

Class: edge_crack (confidence 83.8%)

The visible defect in this semiconduction inspection image is characterized by: 1. Defect Morphology - Edge Crack This type of defect typically appears as a thin crack or break along one or more edges of the die. It can be caused by various factors such as thermal stress, mechanical damage during handling, or manufacturing imperfections. In this case, it seems to affect both inner and outer edges of the chip. 2. Spatial Distribution - Random Pattern The cracks appear randomly distributed across different areas within the die. There doesn't seem to be any specific pattern or concentration that would indicate a particular cause for these cracks. 3. Size - Microscopic Level Based on the resolution provided, I cannot determine exact dimensions but given their appearance in the image, they likely fall under microscopic level sizes which could range from tens to hundreds of micrometers depending on the magnification used when capturing the image. Typically, microscopic-level features require specialized equipment like scanning electron microscope (SEM) or focused ion beam (FIB) imaging techniques to measure with precision. It should be noted that while DINOv0 confidences above 95% usually signify high reliability, lower confidences may still represent valid observations if other forms of evidence support them. However, without additional context about what constitutes "high" vs "low" confidences in this system, we must rely solely on the visual information presented in

Root Cause Analysis

Equipment telemetry (simulated SECS/GEM): SECS/GEM S5F1 Alarm 0x0042: vacuum_level=448 mbar (threshold 500 mbar) at T-45 min. S6F11 CEID 1012: handler_speed=322 mm/s (limit 300 mm/s) at T-30 min. chuck_temp=24.1 C (nominal). bond_force=18.4 g (nominal). Historical: particle_contamination on LOT-2024-031 linked to handler_speed overage.

1. Hypothesis 1: The high vacuum level of 448 mbars observed in the SECS/Gem S5F alarm log might have led to a buildup of particles within the system that contributed to the formation of the crack along the edge of the die. This could be due to insufficient cleaning procedures prior to wafer handling or during processing steps
2. Hypothesis 2: Exceeding the limit for handler speed with a value of 322mm/s as recorded by the CEID

Corrective Actions

1. To address the critical defect identified as "edge_crack" on equipment model EQ-INSPI-01, consider implementing the following three specific corrective measures with associated process parameters targets:

2. Enhance pre-processing cleaning protocols by increasing ultrasonic bath time from current 2 minutes to at least 6 minutes for effective removal of contaminants before wafer handling. Target: Ultrasonic Bath Time = 6 min
3. Optimize post-etching rinses using deionized water (DI) instead of tap water. Replace tap water with DI water immediately after etching processes. Target: Rinse Water Type = Deionized Water
4. Implement regular maintenance checks for the vacuum pump used in the SECM/SEMS chamber. Ensure proper functioning of the vacuum pump by monitoring pressure levels regularly. Set target threshold values for vacuum pressure: Low Pressure Threshold= 99% Vacuum Level and High Pressure Thresholds= 97%. Regularly inspect and replace parts if necessary to maintain optimal performance
5. By addressing these key areas through targeted improvements, it is expected that the frequency of such critical defects can be significantly reduced, ensuring higher quality output from your manufacturing processes